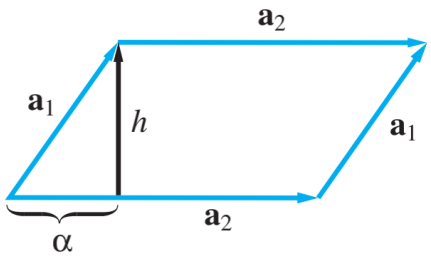
MAT 2040 Linear Algebra

Assignment 7

Please note that

- Released date:
- Due:
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW7".

Question 1. Let $\boldsymbol{A}$ be a 2×2 matrix with linearly independent column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. If $\mathbf{a}_1$ and $\mathbf{a}_2$ are used to form a parallelogram P with altitude h (see the figure), show that $h^2 \|\mathbf{a}_2\|^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - \left(\mathbf{a}_1^T \mathbf{a}_2\right)^2$



Solution

By the Pythagorean Theorem

$$\alpha^2 + h^2 = \|\mathbf{a}_1\|^2$$

where α is the scalar projection of $\mathbf{a}_1$ into $\mathbf{a}_2$. It follows that

$$\alpha^2 = \frac{(\mathbf{a}_1^T \mathbf{a}_2)^2}{\|\mathbf{a}_2\|^2}$$

and

$$h^2 = \|\mathbf{a}_1\|^2 - \frac{(\mathbf{a}_1^T \mathbf{a}_2)^2}{\|\mathbf{a}_2\|^2}$$

Hence

$$h^2 \|\mathbf{a}_2\|^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - \left(\mathbf{a}_1^T \mathbf{a}_2\right)^2$$

Question 2. Find the constants a and b such that the vectors $\mathbf{u} = \begin{bmatrix} a \\ 4 \\ -b \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ 1 \\ b \end{bmatrix}$ are orthogonal and $a = b + 1$.

Solution

Vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, hence their inner product is equal to zero

$$a^2 + 4 - b^2 = 0.$$

Given that $a = b + 1$, substitute a by $b + 1$ in the above equation

$$(b+1)^2 + 4 - b^2 = 0.$$

Expand the above equation and simplify

$$2b + 5 = 0.$$

Solve for b

$$b = -\frac{5}{2}.$$

Find a

$$a = b + 1 = -\frac{5}{2} + 1 = -\frac{3}{2}$$

Hence, the vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $a = -\frac{3}{2}$ and $b = -\frac{5}{2}$.

Question 3. State whether each of the following is true or false:

(a) If the subspaces V and W are orthogonal, then $V^\perp$ and $W^\perp$ are also orthogonal.

(b) If V is orthogonal to W and W orthogonal to Z then V is orthogonal to Z.

Solution

(a) In $\mathbb{R}^n$, suppose that $V = W = \{0\}$. Then since the zero vector is orthogonal to itself, V is orthogonal to $W = V$. However we have $V^\perp = W^\perp = \mathbb{R}^n$ and $\mathbb{R}^n$ is not orthogonal to itself, so $V^\perp$ and $W^\perp$ are not orthogonal. The statement is false.
(b) Suppose that $V = Z = \mathbb{R}^n$ and $W = \{0\}$. Then V is orthogonal to W and W is orthogonal to Z but V is not orthogonal to Z because $\mathbb{R}^n$ is not orthogonal to itself. The statement is false.

Question 4. Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^n$ and let $\mathcal{S} = \text{Span}\{\mathbf{x}, \mathbf{y}\}$. Construct a matrix by $\boldsymbol{A} = \mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$.

(a) Show that $\boldsymbol{A}$ is symmetric.

(b) Show that $\text{Null}(\boldsymbol{A}) = \mathcal{S}^\perp$.

Solution

(a) $\boldsymbol{A}^T = \left(\mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T\right)^T = \left(\mathbf{x}\mathbf{y}^T\right)^T + \left(\mathbf{y}\mathbf{x}^T\right)^T = \mathbf{y}\mathbf{x}^T + \mathbf{x}\mathbf{y}^T = \boldsymbol{A}$.
(b) For any vector $\mathbf{z} \in \mathbb{R}^n$,

$$\boldsymbol{A}\mathbf{z} = \mathbf{x}\mathbf{y}^T\mathbf{z} + \mathbf{y}\mathbf{x}^T\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y},$$

where $c_1 = \mathbf{y}^T\mathbf{z}$ and $c_2 = \mathbf{x}^T\mathbf{z}$. If $\mathbf{z} \in \text{Null}(\boldsymbol{A})$,

$$\mathbf{0} = \boldsymbol{A}\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y}$$

and since $\mathbf{x}$ and $\mathbf{y}$ are linearly independent, we have $c_1 = \mathbf{y}^T\mathbf{z} = \mathbf{0}$ and $c_2 = \mathbf{x}^T\mathbf{z} = 0$. So $\mathbf{z} \perp \mathbf{x}, \mathbf{z} \perp \mathbf{y} \Rightarrow \mathbf{z} \in \mathcal{S}^\perp$. Conversely, if $\mathbf{z} \in \mathcal{S}^\perp$, it follows that

$$\boldsymbol{A}\mathbf{z} = c_1\mathbf{x} + c_2\mathbf{y} = \mathbf{0}.$$

Therefore, $\mathbf{z} \in \text{Null}(\boldsymbol{A})$.

Question 5. Find the angle between the vectors $\mathbf{v}$ and $\mathbf{w}$:

$$\mathbf{v} = (2, 1, 3)^T, \ \mathbf{w} = (6, 3, 9)^T$$

Solution

The angle θ between two nonzero vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$ satisfies

$$\begin{aligned}\cos\theta &= \frac{\mathbf{v}^T\mathbf{w}}{\|\mathbf{v}\|\|\mathbf{w}\|}, \quad 0 \le \theta \le \pi \\ &= 1\end{aligned}$$

So $\theta = 0°$.

Question 6. For the following pair of vectors $\mathbf{x}$ and $\mathbf{y}$, find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$ and verify that $\mathbf{p}$ and $\mathbf{x} - \mathbf{p}$ are orthogonal:

$$\mathbf{x} = (3,5)^T,\ \mathbf{y} = (1,1)^T$$

Solution

The vector projection of $\mathbf{x}$ onto $\mathbf{y}$ is

$$\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y} = (4,4)^T.$$

Therefore, $\mathbf{x} - \mathbf{p} = (3,5)^T - (4,4)^T = (-1,1)^T$. It is easy to verify that

$$\mathbf{p}^T(\mathbf{x} - \mathbf{p}) = 0 \quad \Rightarrow \ \mathbf{p} \text{ and } \mathbf{x} - \mathbf{p} \text{ are orthogonal.}$$

Question 7. If $\mathbf{x} = (x_1, x_2)^T, \mathbf{y} = (y_1, y_2)^T$, and $\mathbf{z} = (z_1, z_2)^T$ are arbitrary vectors in $\mathbb{R}^2$, prove that:

(a) $\mathbf{x}^T\mathbf{x} \ge 0$

(b) $\mathbf{x}^T\mathbf{y} = \mathbf{y}^T\mathbf{x}$

(c) $\mathbf{x}^T(\mathbf{y} + \mathbf{z}) = \mathbf{x}^T\mathbf{y} + \mathbf{x}^T\mathbf{z}$

Solution

(a) $\mathbf{x}^T\mathbf{x} = x_1^2 + x_2^2 \ge 0$

(b) $\mathbf{x}^T\mathbf{y} = x_1y_1 + x_2y_2 = \mathbf{y}^T\mathbf{x}$

(c) $\mathbf{x}^T(\mathbf{y} + \mathbf{z}) = x_1y_1 + x_1z_1 + x_2y_2 + x_2z_2 = \mathbf{x}^T\mathbf{y} + \mathbf{x}^T\mathbf{z}$

Question 8. (a) Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{x} = (x_1, x_2, x_3)^T$ and $\mathbf{y} = (y_1, y_2, y_3)^T$. Let

$$\boldsymbol{A} = \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{pmatrix}.$$

Show that $S^{\perp} = \text{Null}(\boldsymbol{A})$.

(b) Find the orthogonal complement of the subspace of $\mathbb{R}^3$ spanned by $(1, 2, 1)^T$ and $(1, -1, 2)^T$.

Solution

(a) Let $(x, y, z)^T \in S^{\perp}$. By the definition of orthogonal complement, we have

$$\begin{cases} \mathbf{x}^T(x, y, z)^T = 0 \\ \mathbf{y}^T(x, y, z)^T = 0 \end{cases} \Rightarrow \begin{cases} x_1 x + x_2 y + x_3 z = 0 \\ y_1 x + y_2 y + y_3 z = 0 \end{cases}$$

This system can be written in a matrix form, i.e.,

$$\begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0,$$

which implies $(x, y, z)^T \in \text{Null}(\boldsymbol{A})$. Thus,

$$(x, y, z)^T \in S^{\perp} \;\Rightarrow\; (x, y, z)^T \in \text{Null}(\boldsymbol{A}).$$

Conversely, we have

$$(x, y, z)^T \in \text{Null}(\boldsymbol{A}) \;\Rightarrow\; (x, y, z)^T \in S^{\perp}.$$

Therefore, $S^{\perp} = \text{Null}(\boldsymbol{A})$.

(b) From part (a), we have the orthogonal component of the subspace S is $S^{\perp} = \text{Null}(\boldsymbol{A})$, where

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix}.$$

To find the null space of $\boldsymbol{A}$, we first reduce the matrix into echelon form, i.e.,

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \end{bmatrix} \overset{R_2 \to R_2 - R_1}{\longrightarrow} \begin{bmatrix} 1 & 2 & 1 \\ 0 & -3 & 1 \end{bmatrix}.$$

Therefore, any $\mathbf{x} = [x_1, x_2, x_3]^T \in \text{Null}(\boldsymbol{A})$ satisfies:

$$\begin{cases} x_1 + 2x_2 + x_3 = 0 \\ -3x_2 + x_3 = 0 \end{cases} \Rightarrow \begin{cases} x_1 + 2x_2 + x_3 = 0 \\ x_3 = 3x_2 \end{cases}.$$

Suppose that $x_2 = t, t \in \mathbb{R}$, then, $x_3 = 3t$. Putting $x_2 = t, x_3 = 3t$ in the equation $x_1 + 2x_2 + x_3 = 0$, we have $x_1 = -5t$.

Therefore, the null space of $\boldsymbol{A}$ is,

$$\begin{aligned} \text{Null}(\boldsymbol{A}) &= \left\{(x_1, x_2, x_3)^T\right\} \\ &= \left\{(-5t, t, 3t)^T,\ t \in \mathbb{R}\right\} \\ &= \left\{(-5, 1, 3)^T t,\ t \in \mathbb{R}\right\} \end{aligned}$$

Question 9. For each of the following systems $\boldsymbol{A}\mathbf{x} = \mathbf{b}$, find all least squares solutions:

(a) $\boldsymbol{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

(b) $\boldsymbol{A} = \begin{pmatrix} 1 & 1 & 3 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \\ 8 \end{pmatrix}$

Solution

Definition: If $\boldsymbol{A}$ is an $m \times n$ matrix of rank n, the normal equations

$$\boldsymbol{A}^T\boldsymbol{A}\mathbf{x} = \boldsymbol{A}^T\mathbf{b}$$

have a unique solution

$$\mathbf{x} = \left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\mathbf{b}$$

and $\mathbf{x}$ is the unique least squares solution of the system $\boldsymbol{A}\mathbf{x} = \mathbf{b}$.

(a) Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ Substitute the values of $\boldsymbol{A}, \mathbf{x}$, and $\mathbf{b}$ in $\boldsymbol{A}^T\boldsymbol{A}\mathbf{x} = \boldsymbol{A}^T\mathbf{b}$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}^T \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{bmatrix}^T \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

When solving this linear system, we can obtain

$$x_1 + 2x_2 = 1$$

Note that there are two variables (namely, x_1, x_2) and one equation. So, solution of it must contain 1 variable as free variable. Suppose x_2 be the free variable. That is, $x_2 = \alpha, \alpha \in \mathbb{R}$ Then from the equation $x_1 + 2x_2 = 1$ get $x_1 = 1 - 2\alpha$ Hence the least squares solution is, $(x_1, x_2)^T = \left\{(1 - 2\alpha, \alpha)^T \mid \alpha \in \mathbb{R}\right\}$.

(b) Similar with (a), we can solve the linear system to obtain:

$$x_1 + 2x_3 = 2$$
$$x_2 + x_3 = 1$$

Note that there are three variables (namely, x_1, x_2, x_3) and two equations. So, solution of it must contains 1 variable as free variable. Suppose x_3 be the free variable. That is, $x_3 = \alpha, \alpha \in \mathbb{R}$ Then from the equation $x_2 + x_3 = 1$ get $x_2 = 1 - \alpha$ from the equation $x_1 + 2x_3 = 2$ get $x_1 = 2 - 2\alpha$ Hence the least squares solution is

$$(x_1, x_2, x_3)^T = \left\{(2 - 2\alpha, 1 - \alpha, \alpha)^T \mid \alpha \in \mathbb{R}\right\}$$

Question 10. Let $\boldsymbol{P} = \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T$, where $\boldsymbol{A}$ is an $m \times n$ matrix of rank n.

(a) Show that $\boldsymbol{P}^2 = \boldsymbol{P}$.

(b) Prove that $\boldsymbol{P}^k = \boldsymbol{P}$ for $k = 1, 2, \ldots$

(c) Show that $\boldsymbol{P}$ is symmetric. [Hint: If $\boldsymbol{B}$ is non-singular, then $\left(\boldsymbol{B}^{-1}\right)^T = \left(\boldsymbol{B}^T\right)^{-1}$.]

Solution

(a)

$$\begin{aligned}
\boldsymbol{P}^2 &= \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \cdot \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \\
\boldsymbol{P}^2 &= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\left(\boldsymbol{A}^T\boldsymbol{A}\right)\right]\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \\
\boldsymbol{P}^2 &= \boldsymbol{A}\left[(\boldsymbol{B})^{-1}(\boldsymbol{B})\right]\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}A^T \quad \left(\text{let } \boldsymbol{B} = \boldsymbol{A}^T\boldsymbol{A}\right) \\
\boldsymbol{P}^2 &= \boldsymbol{A}[\boldsymbol{I}]\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \quad \left(\text{ Since } \boldsymbol{B}\boldsymbol{B}^{-1} = \boldsymbol{I}\right) \\
\boldsymbol{P}^2 &= \boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T \quad (\text{ Since } \boldsymbol{A}\cdot\boldsymbol{I} = \boldsymbol{A}) \\
\boldsymbol{P}^2 &= \boldsymbol{P}
\end{aligned}$$

(b) The objective is to prove that $\boldsymbol{P}^k = P$, for $k = 1, 2, \ldots$ We prove this by using induction on k. If $k = 1$ then we have $\boldsymbol{P}^1 = P$ Assume $\boldsymbol{P}^m = \boldsymbol{P}$, for some m. Consider

$$\begin{aligned}
\boldsymbol{P}^{m+1} &= \boldsymbol{P}\cdot\boldsymbol{P}^m \\
&= \boldsymbol{P}\cdot\boldsymbol{P} \\
&= \boldsymbol{P}^2 \\
&= \boldsymbol{P}
\end{aligned}$$

Hence the result is true by induction on k.

(c) The objective is to show that $\boldsymbol{P}$ is symmetric. It is enough to prove $\boldsymbol{P}^T = \boldsymbol{P}$. Consider

$$\begin{aligned}
\boldsymbol{P}^T &= \left[\boldsymbol{A}\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\boldsymbol{A}^T\right]^T \\
&= \left(\boldsymbol{A}^T\right)^T\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)^{-1}\right]^T\boldsymbol{A}^T \\
&= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)^T\right]^{-1}\boldsymbol{A}^T \quad \left(\begin{array}{l}\text{Here } \boldsymbol{A} \text{ is } m\times n \text{ matrix rank } n, \text{ so} \\ \boldsymbol{A}^T\boldsymbol{A} \text{ is non-singular matrix}\end{array}\right) \\
&= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\left(\boldsymbol{A}^T\right)^T\right)\right]^{-1}\boldsymbol{A}^T \\
&= \boldsymbol{A}\left[\left(\boldsymbol{A}^T\boldsymbol{A}\right)\right]^{-1}\boldsymbol{A}^T \\
&= \boldsymbol{P}
\end{aligned}$$

Therefore, $\boldsymbol{P}$ is symmetric.

Question 11. Let $\mathbf{x} = (1,1,1,1)^T$ and $\mathbf{y} = (8,2,2,0)^T$.

(a) Determine the angle θ between $\mathbf{x}$ and $\mathbf{y}$.

(b) Find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$.

(c) Verify that $\mathbf{x} - \mathbf{p}$ is orthogonal to $\mathbf{p}$.

(d) Compute $\|\mathbf{x} - \mathbf{p}\|, \|\mathbf{p}\|, \|\mathbf{x}\|$ and verify that the Pythagorean law is satisfied.

Solution

(a) we know that $\cos\theta = \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|} = \frac{12}{\sqrt{4}\sqrt{72}} = \frac{6}{\sqrt{72}} = \frac{6}{2\sqrt{6}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = \pi/4$

(b) The vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$ is

$$\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y} = \frac{12}{72}(8,2,2,0)^T = \left(\frac{4}{3}, \frac{1}{3}, \frac{1}{3}, 0\right)^T$$

(c)

$$\mathbf{x} - \mathbf{p} = (1,1,1,1)^T - \left(\frac{4}{3}, \frac{1}{3}, \frac{1}{3}, 0\right)^T = \left(\frac{-1}{3}, \frac{2}{3}, \frac{2}{3}, 1\right)^T$$

$$(\mathbf{x} - \mathbf{p})^T\mathbf{p} = \frac{-4}{9} + \frac{2}{9} + \frac{2}{9} + 0 = 0$$

(d)

$$\mathbf{x} - \mathbf{p} = \left(\frac{-1}{3}, \frac{2}{3}, \frac{2}{3}, 1\right)^T, \mathbf{p} = \left(\frac{4}{3}, \frac{1}{3}, \frac{1}{3}, 0\right)^T$$

$$\|\mathbf{x} - \mathbf{p}\| = \sqrt{\frac{1}{9} + \frac{4}{9} + \frac{4}{9} + 1} = \sqrt{1+1} = \sqrt{2}$$

$$\|\mathbf{p}\| = \sqrt{\frac{16}{9} + \frac{1}{9} + \frac{1}{9} + 0} = \sqrt{2}$$

$$\|\mathbf{x}\| = \sqrt{1+1+1+1} = \sqrt{4} = 2$$

$$\text{Clearly} \quad \|\mathbf{x} - \mathbf{p}\|^2 + \|\mathbf{p}\|^2 = 4 = \|\mathbf{x}\|^2$$

Pythagorean law is satisfied.

Question 12. Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^2$. If $\|\mathbf{x}\| = 2$ and $\|\mathbf{y}\| = 3$, what, if anything, can we conclude about the possible values of $\left|\mathbf{x}^T\mathbf{y}\right|$?

Solution

If $\mathbf{x}$ and $\mathbf{y}$ are linearly independent and θ is the angle between the vectors, then $|\cos\theta| < 1$ and hence

$$\left|\mathbf{x}^T\mathbf{y}\right| = \|\mathbf{x}\|\|\mathbf{y}\||\cos\theta| \leq 6$$

Question 13. For the following matrix, determine a basis for each of the subspaces $\text{Row}\left(A^T\right)$, $\text{Null}(A), \text{Row}(A)$, and $\text{Null}\left(A^T\right)$:

$$A = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}$$

Solution

- We can find bases for $\text{Row}(A)$ and $\text{Null}(A)$ by transforming A into reduced row echelon form:

$$\begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix} \to \begin{pmatrix} 3 & 4 \\ 0 & 0 \end{pmatrix}$$

  So $\{3, 4\}^T$ forms a basis for the row space of A. if $x \in \text{Null}(A)$, it follows that:

$$3\text{x}_1 + 4\text{x}_2 = 0$$

  Thus $(-4, 3)^T$ forms a basis for $\text{Null}(A)$

- We can also find bases for $\text{Row}\left(A^T\right)$ and $\text{Null}\left(A^T\right)$ by transforming A^T into reduced row echelon form:

$$\begin{pmatrix} 3 & 6 \\ 4 & 8 \end{pmatrix} \to \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$$

  Thus, $(1, 2)^T$ forms a basis for the row space of A^T. if $x \in \text{Null}\left(A^T\right)$, it follows that:

$$\mathbf{x}_1 + 2\mathbf{x}_2 = 0$$

  Thus $(-2, 1)^T$ forms a basis for $\text{Null}\left(A^T\right)$

- Conclude:

  $\left\{(3, 4)^T\right\}$: basis for $\text{Row}(A)$

  $\left\{(-4, 3)^T\right\}$: basis for $\text{Null}(A)$

  $\left\{(1, 2)^T\right\}$: basis for $\text{Row}\left(A^T\right)$,

  $\left\{(-2, 1)^T\right\}$: basis for $\text{Null}\left(A^T\right)$

Question 14. Let

$$\mathbf{x} = \begin{pmatrix} 4 \\ 4 \\ -4 \\ 4 \end{pmatrix} \qquad \text{and} \qquad \mathbf{y} = \begin{pmatrix} 4 \\ 2 \\ 2 \\ 1 \end{pmatrix}$$

(a) Determine the angle between $\mathbf{x}$ and $\mathbf{y}$.

(b) Determine the distance between $\mathbf{x}$ and $\mathbf{y}$.

Solution

(a) To find the angle between two vectors $\mathbf{x}$ and $\mathbf{y}$, we can use the dot product formula:

$$\cos\theta = \frac{\mathbf{x}^T\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$

Using the given vectors, we have:

$$\begin{aligned}\mathbf{x}^T\mathbf{y} &= (4)(4) + (4)(2) + (-4)(2) + (4)(1) = 20\\ \|\mathbf{x}\| &= \sqrt{4^2 + 4^2 + (-4)^2 + 4^2} = \sqrt{64} = 8\\ \|\mathbf{y}\| &= \sqrt{4^2 + 2^2 + 2^2 + 1^2} = \sqrt{25} = 5\end{aligned}$$

Substituting these values into the formula for $\cos\theta$, we get:

$$\cos\theta = \frac{20}{(8)(5)} = \frac{1}{2}$$

Taking the inverse cosine of both sides, we get:

$$\theta = \cos^{-1}\left(\frac{1}{2}\right) \approx 60^\circ$$

Therefore, the angle between $\mathbf{x}$ and $\mathbf{y}$ is approximately 60°.

(b)

$$\|\mathbf{x} - \mathbf{y}\| = \sqrt{0^2 + 2^2 + (-6)^2 + 3^2} = \sqrt{49} = 7$$

Therefore, the distance between $\mathbf{x}$ and $\mathbf{y}$ is 7.

Question 15. Let $\mathbf{x}$ and $\mathbf{y}$ be vectors in $\mathbb{R}^n$ and define

$$\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y} \quad \text{and} \quad \mathbf{z} = \mathbf{x} - \mathbf{p}$$

(a) Show that $\mathbf{p} \perp \mathbf{z}$. Thus, $\mathbf{p}$ is the vector projection of $\mathbf{x}$ onto $\mathbf{y}$; that is, $\mathbf{x} = \mathbf{p}+\mathbf{z}$, where $\mathbf{p}$ and $\mathbf{z}$ are orthogonal components of $\mathbf{x}$, and $\mathbf{p}$ is a scalar multiple of $\mathbf{y}$.

(b) If $\|\mathbf{p}\| = 6$ and $\|\mathbf{z}\| = 8$, determine the value of $\|\mathbf{x}\|$.

Solution

(a) Let $\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}$ and $\mathbf{z} = \mathbf{x} - \mathbf{p}$. We want to show that $\mathbf{p} \perp \mathbf{z}$, that is, $\mathbf{p}^T\mathbf{z} = 0$. We have $\mathbf{p}^T\mathbf{z} = \mathbf{p}^T(\mathbf{x} - \mathbf{p}) = \mathbf{p}^T\mathbf{x} - \mathbf{p}^T\mathbf{p}$.

Substituting the definition of $\mathbf{p}$, we get

$$\mathbf{p}^T\mathbf{x} = \left(\frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}\right)^T \mathbf{x} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}^T\mathbf{x} = \frac{\left(\mathbf{x}^T\mathbf{y}\right)^2}{\mathbf{y}^T\mathbf{y}}$$

Similarly, we can compute $\mathbf{p}^T\mathbf{p}$ as

$$\mathbf{p}^T\mathbf{p} = \left(\frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}\right)^T \left(\frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y}\right) = \frac{\left(\mathbf{x}^T\mathbf{y}\right)^2}{\left(\mathbf{y}^T\mathbf{y}\right)^2}\mathbf{y}^T\mathbf{y} = \frac{\left(\mathbf{x}^T\mathbf{y}\right)^2}{\mathbf{y}^T\mathbf{y}}$$

Thus, we have

$$\mathbf{p}^T\mathbf{z} = 0.$$

(b) We have $\mathbf{x} = \mathbf{p} + \mathbf{z}$, where $\mathbf{p}$ and $\mathbf{z}$ are orthogonal components of $\mathbf{x}$. Therefore, we can use the Pythagorean theorem to write:

$$\|\mathbf{x}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{z}\|^2$$

Substituting the given values, we get:

$$\|x\|^2 = 6^2 + 8^2 = 100$$

Therefore, the value of $|\mathbf{x}|$ is 10.

Question 16. Let S be the subspace of $\mathbb{R}^n$ spanned by the vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$. Show that $\mathbf{y} \in S^\perp$ if and only if $\mathbf{y} \perp \mathbf{x}_i$ for $i = 1, \ldots, k$.

Solution

($\Rightarrow$) Suppose $\mathbf{y} \in S^\perp$. Then, by definition, $\mathbf{y}$ is orthogonal to every vector in S. In particular, $\mathbf{y} \perp \mathbf{x}_i$ for $i = 1, \ldots, k$.

($\Leftarrow$) Now suppose $\mathbf{y} \perp \mathbf{x}_i$ for $i = 1, \ldots, k$. To show that $\mathbf{y} \in S^\perp$, we need to show that $\mathbf{y}$ is orthogonal to every vector in S. Let $\mathbf{x}$ be an arbitrary vector in S, so we can write $\mathbf{x}$ as a linear combination of $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k$, say $\mathbf{x} = a_1\mathbf{x}_1 + a_2\mathbf{x}_2 + \cdots + a_k\mathbf{x}_k$ for some scalars $a_1, a_2, \ldots, a_k$.

Then we have:

$$\mathbf{y}^T\mathbf{x} = a_1\mathbf{y}^T\mathbf{x}_1 + a_2\mathbf{y}^T\mathbf{x}_2 + \cdots + a_k\mathbf{y}^T\mathbf{x}_k = 0$$

where the last equality follows from the fact that $\mathbf{y} \perp \mathbf{x}_i$ for $i = 1, \ldots, k$. Therefore, $\mathbf{y}$ is orthogonal to every vector in S, so $\mathbf{y} \in S^\perp$.

Thus, we have shown that $\mathbf{y} \in S^\perp$ if and only if $\mathbf{y} \perp \mathbf{x}_i$ for $i = 1, \ldots, k$.